

1. Let $C(\alpha, \beta)$ be the circumcentre of the triangle formed by the lines $4x + 3y = 69$, $4y - 3x = 17$, and $x + 7y = 61$. Then $(\alpha - \beta)^2 + \alpha + \beta$ is equal to

[2023 (08 Apr Shift 1)]

- (1) 18
(2) 17
(3) 15
(4) 16

2. If the points P and Q are respectively the circumcenter and the orthocentre of a $\triangle ABC$, then $\vec{PA} + \vec{PB} + \vec{PC}$ is equal to _____

[2023 (10 Apr Shift 2)]

- (1) $2\vec{QP}$
(2) $2\vec{PQ}$
(3) $\vec{PQ}$
(4) $\vec{QP}$

3. In a triangle ABC , if $\cos A + 2 \cos B + \cos C = 2$ and the lengths of the sides opposite to the angles A and C are 3 and 7 respectively, then $\cos A - \cos C$ is equal to

[2023 (12 Apr Shift 1)]

- (1) $\frac{9}{7}$
(2) $\frac{10}{7}$
(3) $\frac{5}{7}$
(4) $\frac{3}{7}$

4. Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C . If the orthocenter of the triangle ABC is $(\alpha, \beta, \frac{6}{7})$, then $98(\alpha + \beta)^2$ is equal to _____.

[2023 (12 Apr Shift 1)]

5. Let for a triangle ABC

$$\vec{AB} = -2\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$$

$$\vec{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$$

If $\delta > 0$ and the area of the triangle ABC is $5\sqrt{6}$ then $\vec{CB} \cdot \vec{CA}$ is equal to

[2023 (13 Apr Shift 2)]

- (1) 60
(2) 54
(3) 108
(4) 120

ANSWER KEYS

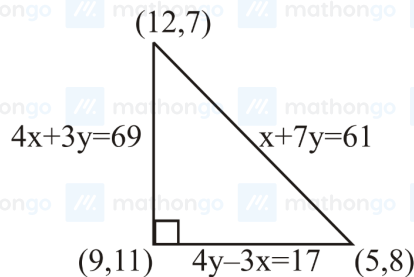
1. (2) 2. (3) 3. (2) 4. (288) 5. (1)

1. (2)

Given,

$C(\alpha, \beta)$ be the circumcentre of the triangle formed by the lines $4x + 3y = 69$, $4y - 3x = 17$, and $x + 7y = 61$,

Now plotting the diagram of the above equations and finding the intersection point we get,



Now from diagram we can see that it is a right-angled triangle, so the circumcentre is given by midpoint of the hypotenuse,

Hence, circumcentre will be, $C\left(\frac{12+5}{2}, \frac{7+8}{2}\right)$

$$\Rightarrow C\left(\frac{17}{2}, \frac{15}{2}\right) = (\alpha, \beta)$$

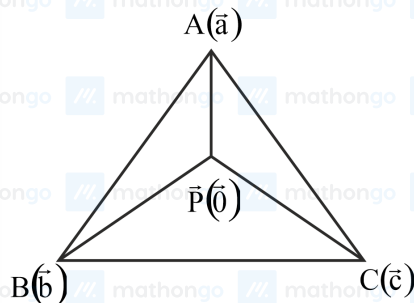
Hence, $(\alpha - \beta)^2 + \alpha + \beta$

$$= \left(\frac{17}{2} - \frac{15}{2}\right)^2 + \frac{17}{2} + \frac{15}{2}$$

$$= 1 + 16 = 17$$

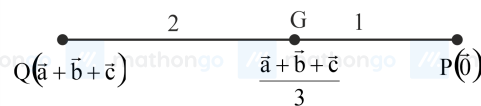
2. (3)

Let P be origin.



Now from the above triangle we can write $\vec{PA} = \vec{a}$, $\vec{PB} = \vec{b}$, $\vec{PC} = \vec{c}$.

Also, we know that centroid G divides orthocentre Q and Circumcentre P in the ratio 2 : 1.



$$\text{Now } \vec{PG} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

$$\Rightarrow \vec{PQ} = 3\left(\frac{\vec{a} + \vec{b} + \vec{c}}{3}\right)$$

$$\Rightarrow \vec{PQ} = \vec{a} + \vec{b} + \vec{c} = \vec{PA} + \vec{PB} + \vec{PC}$$

Therefore, the required answer is $\vec{PQ}$.

3. (2)

Given,

$$AB = 7, BC = 3.$$

$$\Rightarrow c = 7, a = 3.$$

Apply cosine rule in the given equation $\cos A + 2 \cos B + \cos C = 2$, we get

$$\Rightarrow \left(\frac{b^2 + c^2 - a^2}{2bc} \right) + 2 \left(\frac{c^2 + a^2 - b^2}{2ac} \right) + \left(\frac{a^2 + b^2 - c^2}{2ab} \right) = 2$$

$$\Rightarrow \left(\frac{b^2 + 49 - 9}{14b} \right) + \left(\frac{49 + 9 - b^2}{21} \right) + \left(\frac{9 + b^2 - 49}{6b} \right) = 2$$

$$\Rightarrow \left(\frac{b^2 + 40}{14b} \right) + \left(\frac{58 - b^2}{21} \right) + \left(\frac{b^2 - 40}{6b} \right) = 2$$

$$\Rightarrow b^3 - 5b^2 - 16b + 80 = 0$$

$$\Rightarrow (b - 4)(b + 4)(b - 5) = 0$$

$$\Rightarrow b = -4, 4, 5.$$

But $b \neq -4$.

Also, if $b = 4$ triangle cannot be constructed.

That means $b = 5$.

$$\Rightarrow \cos A - \cos C = \left(\frac{b^2 + c^2 - a^2}{2bc} \right) - \left(\frac{a^2 + b^2 - c^2}{2ab} \right)$$

$$\Rightarrow \cos A - \cos C = \left(\frac{5^2 + 7^2 - 3^2}{2(5 \times 7)} \right) - \left(\frac{3^2 + 5^2 - 7^2}{2(3 \times 5)} \right)$$

$$\Rightarrow \cos A - \cos C = \left(\frac{65}{70} \right) - \left(\frac{-15}{30} \right) = \frac{195 + 105}{210} = \frac{300}{210}$$

$$\Rightarrow \cos A - \cos C = \frac{10}{7}$$

4. (288)

Given,

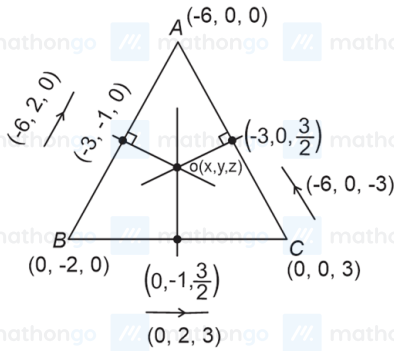
The plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C ,

So, $A(-6, 0, 0), B(0, -2, 0)$ & $C(0, 0, 3)$

And the orthocenter of the triangle ABC is $(\alpha, \beta, \frac{6}{7})$.

We know that, circumcentre of triangle is point of intersection of perpendicular bisectors,

So, plotting the diagram we get,



Now from diagram we can see that O to midpoint of $BC \perp BC$

So, by perpendicular vector formula we get,

$$(x - 0) \times 0 + (y + 1) \times 2 + \left(z - \frac{3}{2}\right) \times 3 = 0$$

$$\Rightarrow 4y + 6z - 5 = 0 \dots\dots(1)$$

Similarly, for side AC we get,

$$(x + 3) \times (-6) + (y - 0) \times 0 + \left(z - \frac{3}{2}\right) \times (-3) = 0$$

$$\Rightarrow 4x + 2z + 9 = 0 \dots\dots(2)$$

Similarly, for side AB we get,

$$\Rightarrow (x + 3) \times (-6) + (y + 1) \times 2 + (z - 0) \times 0 = 0$$

$$\Rightarrow 3x - y + 8 = 0 \dots\dots(3)$$

Now from equation (1), (2) & (3) we get,

$$x = \frac{-9}{4} - \frac{z}{2}, y = \frac{5}{4} - \frac{3}{2}z, z = z$$

Now using the relation between centroid, circumcentre and orthocentre, we get,

$$\begin{array}{ccc} \text{O} & \text{G} & \text{H} \\ \left(-\frac{9}{4} - \frac{z}{2}, \frac{5}{4} - \frac{3}{2}z, z\right) & \left(-2, -\frac{2}{3}, 1\right) & \left(\alpha, \beta, \frac{6}{7}\right) \end{array}$$

$$\Rightarrow G = \left(\frac{\alpha - \frac{9}{4} - \frac{z}{2}}{3}, \frac{\beta + \frac{5}{4} - \frac{3}{2}z}{3}, \frac{\frac{6}{7} + 2z}{3}\right)$$

$$\Rightarrow \left(-2, -\frac{2}{3}, 1\right) = \left(\frac{\alpha - \frac{9}{4} - \frac{z}{2}}{3}, \frac{\beta + \frac{5}{4} - \frac{3}{2}z}{3}, \frac{\frac{6}{7} + 2z}{3}\right)$$

Now on comparing both side we get,

$$\Rightarrow z = \frac{15}{14}, \alpha = -\frac{3}{7}, \beta = -\frac{9}{7}$$

$$\Rightarrow 98(\alpha + \beta)^2 = 98 \times \frac{144}{49} = 288$$

5. (1)

Given,

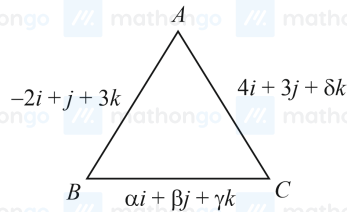
A triangle ABC

$$\vec{AB} = -2\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$$

$$\vec{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$$

Now plotting the diagram we get,



Now from triangle law of addition we get,

$$\vec{CA} + \vec{AB} = \vec{CB}$$

$$\Rightarrow 4\hat{i} + 3\hat{j} + \delta\hat{k} + (-2\hat{i} + \hat{j} + 3\hat{k}) = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$$

$$\Rightarrow 2\hat{i} + 4\hat{j} + (\delta + 3)\hat{k} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$$

Now on comparing both side we get,

$$\alpha = 2, \beta = 4 \text{ \& } \gamma = \delta + 3$$

Now area of triangle is given by,

$$A = \frac{1}{2} |\vec{AB} \times \vec{BC}|$$

$$\Rightarrow 5\sqrt{6} = \left| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & 3 \\ 2 & 4 & \gamma \end{vmatrix} \right|$$

$$\Rightarrow (5\sqrt{6})^2 = (\gamma - 12)^2 + (6 + 2\gamma)^2 + 100$$

$$\Rightarrow 5\gamma^2 = 320$$

$$\Rightarrow \gamma^2 = 64 \Rightarrow \gamma = 8$$

$$\text{So, } \delta = 8 - 3 = 5$$

$$\vec{CB} \cdot \vec{CA} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}) \cdot (4\hat{i} + 3\hat{j} + \delta\hat{k})$$

$$\Rightarrow \vec{CB} \cdot \vec{CA} = (2\hat{i} + 4\hat{j} + 8\hat{k}) \cdot (4\hat{i} + 3\hat{j} + 5\hat{k}) = 8 + 12 + 40 = 60$$